

NAME(S):

Caesar Cipher

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date)

Learning goals / relevant topics

- String methods (Runestone Ch. 7) & ASCII encodings (<https://www.asciitable.com/>)
- Binary to Decimal numbers (<https://www.cuemath.com/numbers/binary-to-decimal/>)

1. Caesar Cipher Encoding

A Caesar cipher (or shift cipher) encrypts text by “shifting” each letter a given number of positions in the alphabet. For example, if the number is 3, then an a would become a d, b would become e and so on. At the end of the alphabet, we “loop back” to the beginning, so a z would become a c.

For example, if we encode the message “Caesar Ciphers are cool” with a shift of 7, it becomes:

Jhlzhy Jpwolyz hyl jvvs.

If we encode that same message with a shift of 1, it becomes:

Dbftbs Djqifst bsf dppm.

1. Download the starter file `caesar.py` from the course website.
2. Write a function called `encipher` that takes a string representing an unencoded message and a number indicating by how many positions each letter should be shifted. The function should return a new string which represents the corresponding Caesar Cipher encoding.

For this problem, you should assume that upper-case letters are encoded as upper-case letters, and all non-alphabetic characters are left unchanged. Some examples:

```
>>> encipher('xyza', 1)
'yzab'
>>> encipher('Z A', 1)
'A B'
>>> encipher("*ab?", 1)
'*bc?'
>>> encipher("This is a string!", 2)
'Vjku ku c uvtkpi!'
>>> encipher("Caesar cipher? I prefer Caesar salad.", 25)
'Bzdrzq bhogdq? H oqdedq Bzdrzq rzkzc.'
```

Feel free to add additional helper functions if you find it useful to do so.

2. Caesar Cipher Decoding:

3. Write a function called `decipher` that takes a message encrypted with a Caesar Cipher and decodes it. The function should take two parameters - the encoded message and a number indicating by how many positions each character was shifted - and it should return the decoded message. For example:

```
>>> decipher('Bzdrzq bhogdq? H oqdedq Bzdrzq rzkzc.', 25)
'Caesar cipher? I prefer Caesar salad.'
```

Don't start a new file for this. Add your new function to the code that you already have in `caesar.py`.

Hint: the built-in functions `ord`, `chr`, and `isalpha` may be helpful for this assignment.

Note: For these optional sections, you will need to look up how to read and write files. We won't start talking about this in class for a while yet.

4. (Optional) Create a new function `encipher_file` that takes a string representing the name of a text file and a number indicating by how many positions each letter should be shifted. This function should open the specified file, then call `encipher` on its contents using the numeric parameter given to `encipher_file`. The enciphered contents of the file should be written out to a new file.
5. (Optional) Create a new function `decipher_file` that takes a string representing the name of an enciphered text file and a number indicating by how many positions each letter was shifted. This function should open the specified file, then call `decipher` on its contents using the numeric parameter given to `decipher_file`. The enciphered contents of the file should be written out to a new file.

Prompt: What string methods and/or operators can you find from Runestone and/or Python documentation?